Roll No.:

MID TERM EXAMINATION 2025

Name of the Program: B. Tech Mechanical | Semester: 6th

Name of the Course: Mechanical Vibrations | Course Code: TME 610

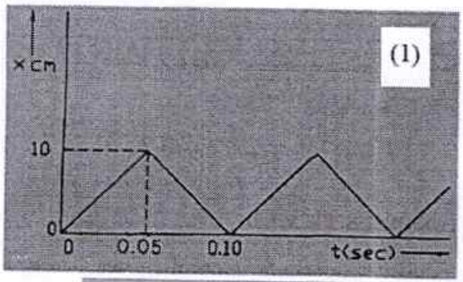
Time: 90 Minutes

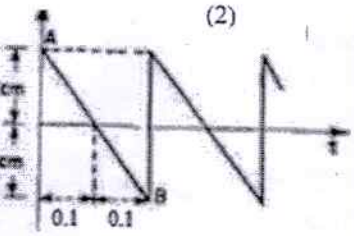
Note:

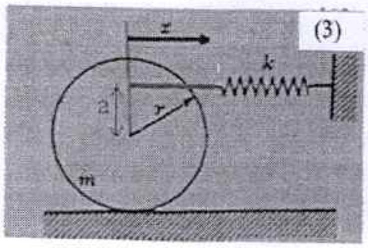
i. Answer all the questions by choosing any one of the sub-questions.

ii. Each question carry 10 marks

Q1	(10 marks)	CO1
(a)	Add the following two motions analytically and check the solution graphically. $x_1 = 3\sin(\omega t+30^\circ)$ $x_2 = 2\cos(\omega t-15^\circ)$	
	OR	
(b)	i. Express the complex number 4+j7 in exponential form. ii. Express the exponential number $5e^{j0.2}$ in complex form.	
Q2	(10 marks)	CO1
(a)	Represent the periodic motion shown in Fig. 1 by Harmonic series	
	OR	
(b)	Represent the periodic motion shown in Fig. 2 by Harmonic series	
Q3	(10 marks)	CO1
(a)	How the displacement, velocity and acceleration vectors of simple harmonic motion $x = X\sin\omega t$ are related to each other. Express by a neat sketch	
	OR	
(b)	Define the following terms related to vibrations i. Cycle ii. Period iii. Cyclic Frequency iv. Angular frequency v. Resonance	
Q4	(10 marks)	CO1 CO2
(a)	Determine the equations of the motions and natural frequency of the system shown in Fig. 3	
	OR	
(b)	Split up the harmonic motion $x = 8\cos(\omega t + \pi/4)$ into two harmonic motions, one of them having phase angle of zero and other having phase angle of 60^0.	
Q5	(10 marks)	
(a)	Determine the equations of the motions and natural frequency of the Simple pendulum mass m, length l using both Newtons and Energy method (Fig. 4)	CO2
	OR	
(b)	Determine the equations of the motions and natural frequency of the pulley with mass M, mass moment of inertia I, radius R, attached with mass m at radius R and spring of stiffness k at radius r. (Fig. 5)	







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